Aldrich Antonio Fernandes

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PROFILE

Robotics software developer with hands-on experience deploying HRI systems on a physical TIAGo robot, competed at RoboCup@Home 2026. Experienced in building end-to-end technical projects spanning machine learning, SaaS systems, and ROS-based robotics, within Linux environments, with a growing interest in robotic automation and applied AI.

EXPERIENCE

Robotics Software Developer & Development Lead *October 2025 – Current*
Learning Autonomous Service Robots (LASR) Team @ KCL

- Helped lead a 10-person engineering team for RoboCup@Home, coordinating task deadlines, onboarding members to the codebase, and contributing key components to the team's robotics software.
- Developed HRI behaviours for the TIAGo robot, including autonomous person-following, receptionist behaviours for guest seating, and bag handover and placement.
- Inherited and extended the General Purpose Service Robot (GPSR) planning logic, addressing execution gaps and implementing autonomous task execution from natural-language commands; deployed, tested, and supervised the system during RoboCup@Home 2026 in Incheon, South Korea.
- Oversaw autonomous robot operation during live competition and KCL NMES Open Days, monitoring execution, troubleshooting failures, and managing rapid recovery and retries under competition time constraints.
- Represented KCL at RoboCup@Home 2026, presenting posters on the team's open-source codebase to international robotics researchers and contributing to the delivery of two key team outputs.

Graduate Teaching Assistant *October 2026 – Ongoing*
King's College London

- Facilitate hands-on robotics labs for Level 5 students, supporting Python-based robotics software development and deployment using ROS and Linux.
- Troubleshoot software and hardware issues across TurtleBot3 Waffle Pi platforms, supporting multiple students during practical experiments and helping ensure smooth lab operation.

Sales Assistant *September 2024 – Ongoing*
Screwfix

- Built rapport with regular customers by identifying patterns in issues and applying proven solutions, demonstrating problem-solving and customer focus.
- Supported warehouse operations by organising inventory, restocking efficiently, and maintaining a clean workspace to improve productivity.

King's Undergraduate Research Fellow (KURF) *June 2026 – July 2026*
King's College London

- Completed a funded research under Dr. Matteo Leonetti on the project "Robots for Healthy Ageing: Engineering Socially Assistive AI".
- Designed and implemented Human-Robot Interaction (HRI) behaviours for the PAL Robotics TIAGo platform, deploying machine learning algorithms and LLMs for robust voice command recognition.
- Executed structured research cycles, including literature reviews, experiment planning, and quantitative data analysis on robot success rates and navigation safety for frail populations.

EXTRACURRICULAR

Cadet Staff Sergeant *November 2019 – July 2024*
Combined Cadet Force

- Led, instructed, and mentored younger cadets while representing the contingent at recruitment events, showcasing leadership and teamwork. As well as organised and managed cadet activities, ensuring high standards of discipline and performance during drills and exercises.
- Displayed resilience, adaptability, and clear decision-making under pressure by often commanding training exercises and expeditions, leading a platoon ($\sim$24) of cadets on multiple occasions.

EDUCATION

King's College London *2024 – 2027*

- Artificial Intelligence BSc
- Current Grade Classification: 1st

St Ignatius College *Sept 2017 – June 2024*

- 3 A-Levels + EPQ: Achieved A*, A, A in Maths, Further Maths and Computer Science respectively.
- 9 GCSE (Attained a grade average of 8.2)

PROJECTS

Loan Approval Predictor
Logistic Regression / Python

- Designed and implemented a logistic regression neural network from scratch in Python to predict mortgage approval outcomes.
- Focused on model architecture, numerical optimisation, and data preprocessing without high-level ML frameworks.

Music Recommendation SaaS
Full-Stack / ML / DevOps / Git / Cloud

- Built a fully containerised and deployable SaaS platform integrating computer vision and ML models.
- Used YOLO and BitNet to infer user context and generate real-time music recommendations.
- Emphasised modular design, containerisation, and deployment-ready architecture.

LASR Open-Source Codebase & HRI Software
ROS (Noetic/ Humble) / Python / Open-Source

- Researched and ported the official LASR team codebase to newer software architectures, successfully migrating systems from ROS 1 to ROS 2.
- Developed and integrated Human-Robot Interaction (HRI) behaviours such as computer vision based guest-seating algorithms and object handover and placement on a physical TIAGo robot.
- Tested and deployed a 'natural language instructions to executable actions' pipeline, adding new skills for the robot to execute, enabling it to autonomously plan and execute tasks of varying complexity from diverse inputs.
- Produced competition-ready code deployed internationally at RoboCup, and successfully demonstrated the software in public settings including the KCL Informatics Showcase and Open Days.